WELCOME BACK!

Instructor Introduction & Agenda

- The G-Safe Energy Wheel
  - What is each energy?
- Pre-Task Planning
  - Overview of Pre-Task Planning
  - Responsibilities
- Follow up- etc.

LET'S GET STARTED!

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Learning Objective

By the end of this training, you will be able to better identify hazards for pre-task planning using the energy wheel.

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The Energy Wheel

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What is the Energy Wheel?

Key Features

- Based on scientific research
- Methods that help you to improve upon the things you are already doing
- Back to the basics approach to focus on what matters most

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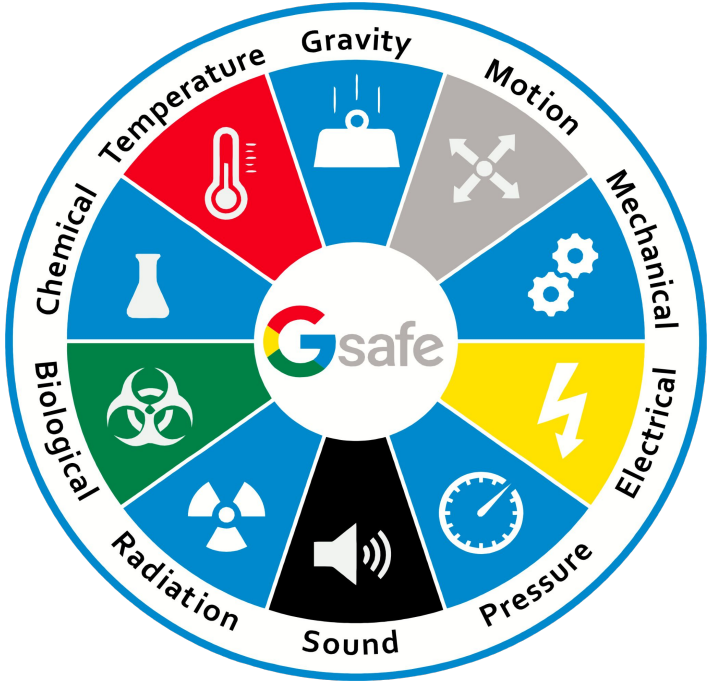
Energy Theory

- Every injury is the result of contact between a human and an energy source
- All relevant energy sources are hazards
- ***So… look for the energy!***
- Research shows that some energy sources are identified by instinct, but we need reminders for others
- The energy wheel helps provide those reminders



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What are the G-Safe Energies?

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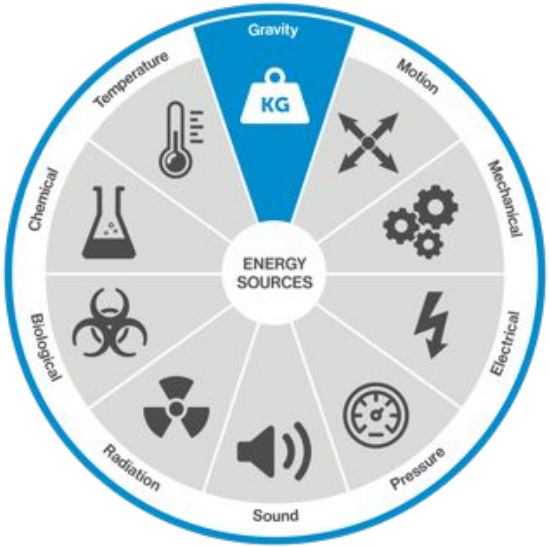
Gravity

Force caused by the attraction of all masses to the mass of the earth

- Uneven work surface
- Work at heights
- Unsecure materials
- Falling objects
- Suspended loads
- Security of materials



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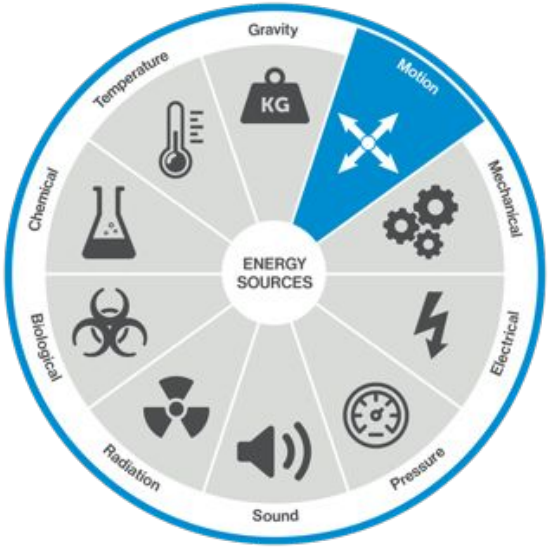
Motion

Change in position of objects or substances

- Sharp objects
- Ergonomics
- Moving equipment or loads
- Flying debris
- Dust particles



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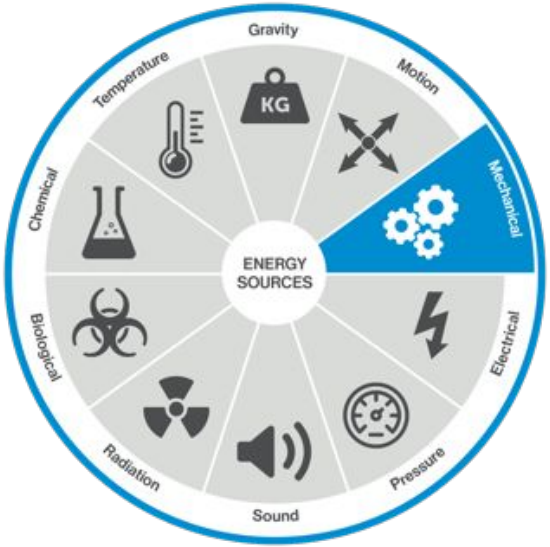
Mechanical

Rotation, vibration, tension, or compression

- Pinch points
- Cables
- Machinery
- Tools
- Vibration



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Mechanical

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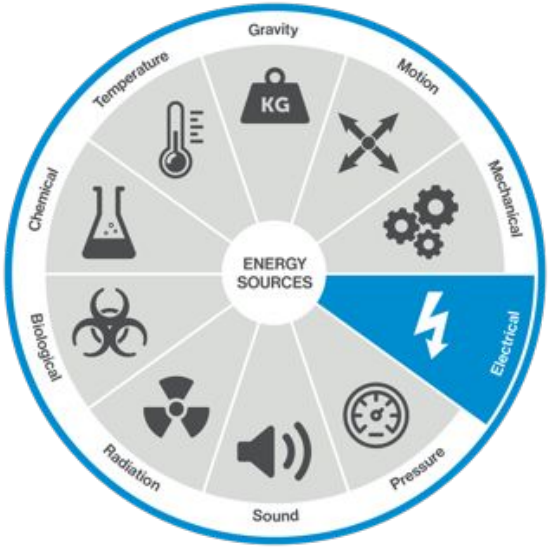
Electrical

The presence of an electrical change or current

- Wires
- Supply
- Power Tools
- Extension cords
- Transformers
- Arc Flash



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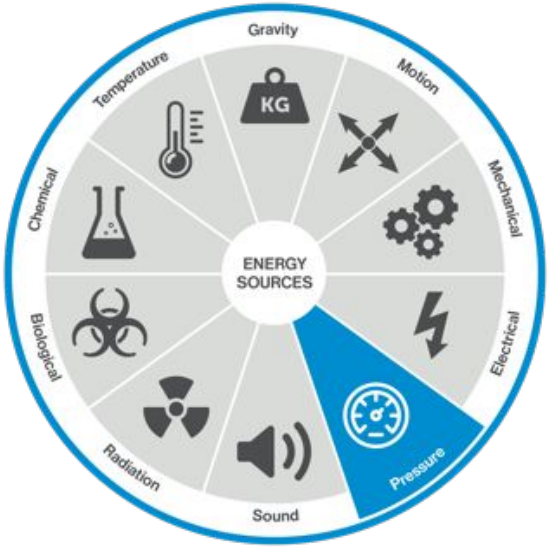
Pressure

Liquid or gas compressed or under a vacuum

- Piping systems
- Hydraulic lines
- Pumps
- Valves
- Closed tanks



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Pressure

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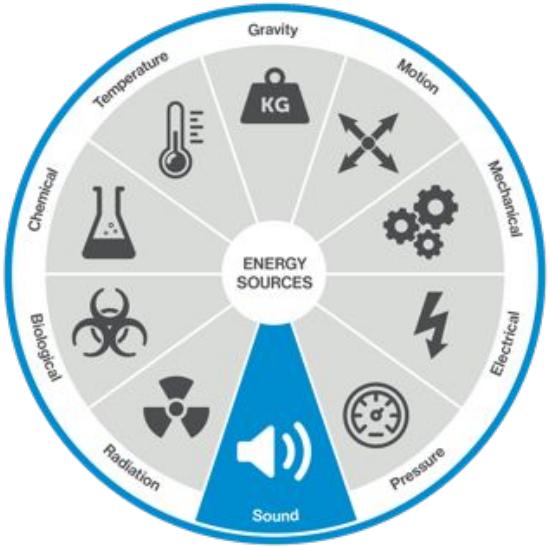
Sound

Audible vibrations caused from the contact of two or more objects

- Heavy machinery
- Power tools
- Engines
- Cutting
- Nail gun
- Hammers



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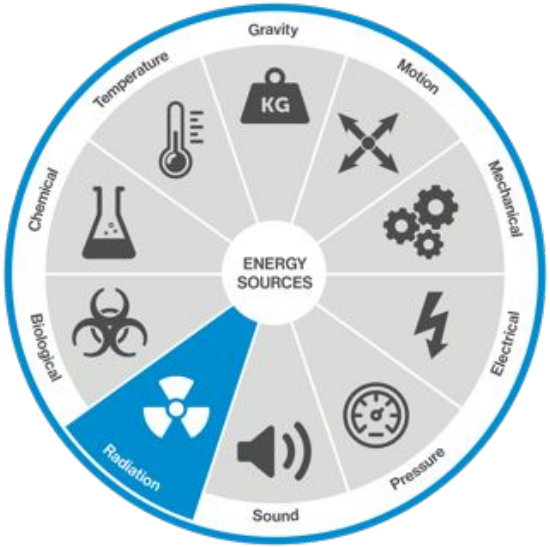
Radiation

Elements that emit ions or atomic particles

- Welding
- Lasers
- Sun exposure
- X-ray testing



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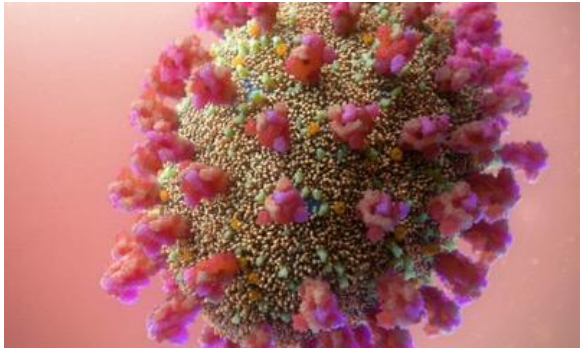
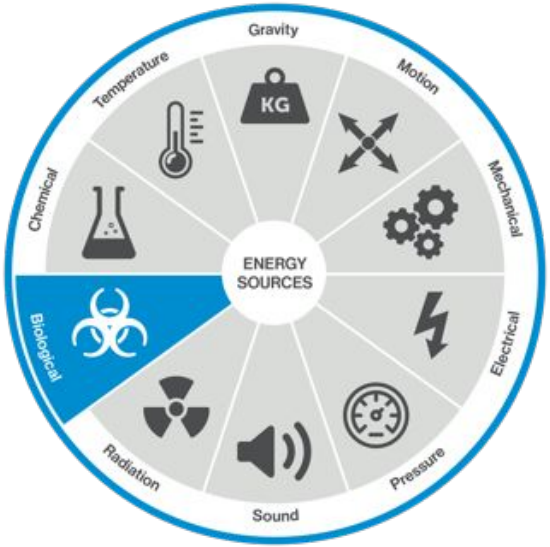
Biological

Living organisms that pose health risks

- Bees
- Mosquitos
- Snakes
- Mold
- Restrooms
- Viruses/COVID





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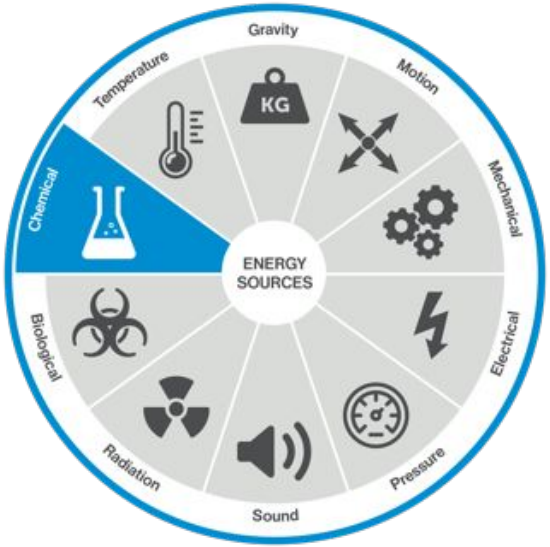
Chemical

Reactive elements in the environment

- Solvents
- Cleaning products
- Engine exhaust
- Dust
- Hazardous waste
- Liquid concrete
- Engulfment



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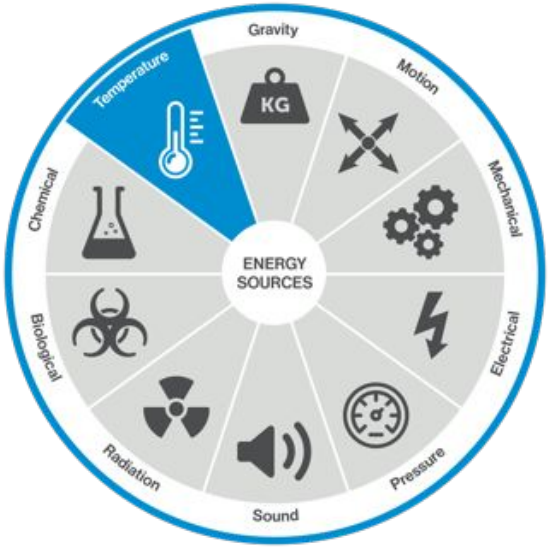
Temperature

Differences in thermal energy with the human body

- Friction
- Engines
- Steam
- Pressure change
- Heat stress



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Pre-Task Planning

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Overview of Pre-Task Planning

- Understand the hazards associated with the planned work for the day
- Complete PTP in the field to maintain understanding of workplace hazards
- Communication for all team members
- Evaluate working area and conditions
  - i.e. walking/working surfaces, excavations, fall hazards, energized systems, congestion, etc.
- Completed at the start of a task and managed throughout as conditions change

"Most injuries occur during task mobilization and demobilization."

Important Considerations

Workplace Barriers

- Communication styles and language barriers
- Experience - years in trade, tools, knowledge
  - Differences in age, background and culture
- Construction industry language (slang)
- Work hours and family hours
- Preference of verbal or electronic communication

Responsibilities

Scope Superintendent

- Ensure team members fully understand the scope of work and expectations of pre-task planning (PTP)
- Frequently check-in with team members on PTP verification

Scope Foreman/Lead

- Complete pre-task cards daily, based on the Integrated Work Plan (IWP) and Job Hazard Analysis (JHA)
- Ensure craft employees understand all criteria completed on the card
- Create crew pre-task card (if applicable)
- Ensure proper materials and tools are available for employees
- Check cards when the task changes
- Read and Sign off on cards and submit at end of shift

Craft Team Member

- Create individual pre-task card (if applicable)
- Ensure each section of card is completed and understood
- Ensure task is understood and recognize hazards in work area
- Ask questions if unsure

Company:
Date:
Task:
Location:
Team Members

Foreman / Lead Signature:

By signing this, I have reviewed the plan and communicated it to my team.

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General Info:

Company/Date/Time/Project/Location needs to be specific

"Card completed by" to be rotated within the crew

Company:
Date:
Task:
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General Info:

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Crew Members:

All team members should sign the PTP before starting work. Larger crews performing the same scope/tasks should split up and complete their own card

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Signature:

Signed by Foreperson/Lead while present at the work location. This is to ensure that crew oversight has approved the work plan after physically seen the risks associated, hazards have been identified and a plan has appropriately been communicated to mitigate the risk

Critical Risk Activities	
☐ Working at Heights	☐ LOTO
☐ Confined Space	☐ Trenching
☐ Heavy Equipment	☐ Crane Lifting

Required Permits	
☐ Hot Work	☐ Live Utility
☐ Dig	☐ Core Drilling
☐ Confined Space	☐ Other

Tools

- ☐ Inspected prior to use
- ☐ All guards / handles in place
- ☐ Proper training completed
- ☐ Properly inspected electrical cords
- ☐ GFCI's for all power tools

Housekeeping

- ☐ All electrical cords out of travel way
- ☐ Trash bin available to clean as you go (never touch trash twice)
- ☐ Area clear of debris and material

Manual Material Handling

- ☐ Handling items over 50 lbs.
- ☐ Mechanical Assist (carts, etc.)
- ☐ Team lifting w/ verbal comm.

Slips, Trips, & Falls

- ☐ Walking paths kept clear
- ☐ Carry nothing that blocks view
- ☐ Holes in work area are identified and remain protected against falls
- ☐ Weather conditions (wet, muddy…)

Equipment

- ☐ Eqpt inspected & documented
- ☐ Operator trained and qualified
- ☐ No using cell phone/ tablet

Fall Protection / Working at Heights

- ☐ Fall protection training complete
- ☐ Fall protection (100% tie-off above 6')
- ☐ All eqpt. / PPE inspected each shift
- ☐ Attached to appropriate anchor point
- ☐ Drop zone below area is barricaded from access
- ☐ All holes are covered, secured, and labeled
- ☐ Tool tethers used
- ☐ Scaffolds / equipment inspected daily
- ☐ Appropriate ladder selected for task

Excavation

- ☐ Ensure excavation is inspected daily & documented
- ☐ Proper slopes / shoring in place
- ☐ Proper barricades installed
- ☐ Proper access / egress
- ☐ Competent Person: _____

Environmental

- ☐ Adequate egress and task lighting
- ☐ Proper precautions for airborne containments (dust / silica, fumes)
- ☐ Drinking water & shade provided
- ☐ All containers properly labeled

Standard PPE + Additional PPE

- ☐ Helmet, safety glasses, high-visibility vests/shirts
- ☐ Gloves (ANSI Cut Level 3)
  - ☐ Cut resistant
  - ☐ Chemical
  - ☐ Impact
- ☐ Face shield (flying debris)
- ☐ Meta-tarsal guards (jack hammering)
- ☐ Respirator
- ☐ Fall protection
- ☐ Hearing protection
- ☐ Goggles
  - ☐ Impact ☐ Chemical
- ☐ Knee pads
- ☐ Kevlar sleeves
- ☐ Arc flash gear
- ☐ Other: _____
- ☐ Other: _____

Lean / 5S

- ☐ Sort: Remove unneeded items
- ☐ Straighten: Place for everything & in place
- ☐ Shine: Clean to detect and correct
- ☐ Standardize: Create daily standards
- ☐ Sustain: Create simple solutions

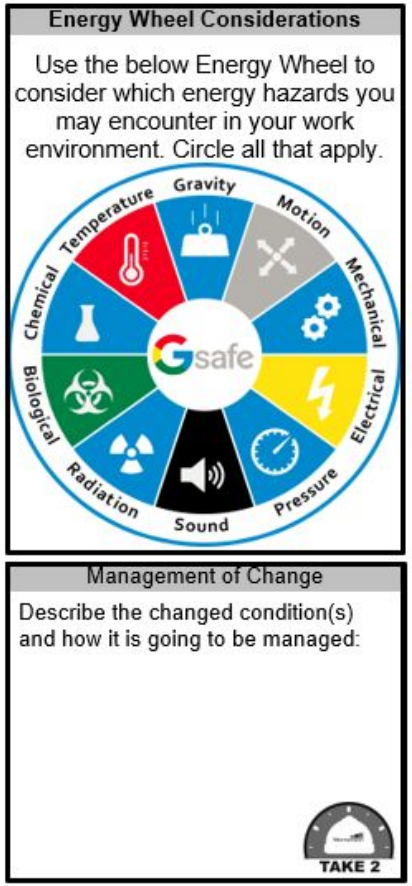
QUALITY

ZERO DEFECTS

Safe Work Principles:

Review this area thoroughly.

Use these safe work principles and G-safe energy wheel as discussion topics and reminders for your crew on how and why the work shall be performed safely



Energy Wheel Considerations:

Determine what energies the crew may encounter throughout the shift

Review each section of the energy wheel to assist in determining any hazards that may have initially been overlooked

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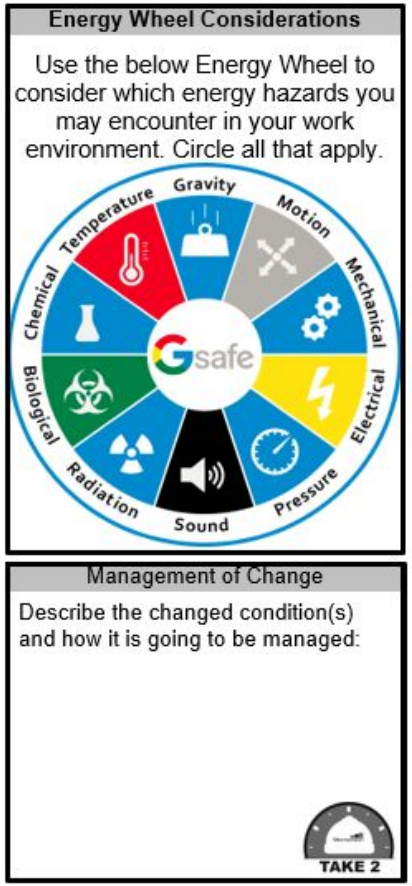

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Determine what energies the crew may encounter throughout the shift

Review each section of the energy wheel to assist in determining any hazards that may have initially been overlooked

Management of Change:

Required - When returning to specific work area to find other trades conducting work that were not part of the original plane developed at beginning of shift = review the plan again and ID hazards

NEW CARD REQUIRED - When starting new / unrelated task that was a change from the original plan developed at beginning of shift

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Steps/Hazards/Control Measures

Steps to Complete Work	Hazards	Control Measures

Identify the various steps needed to complete each task from the tasks identified in section 2.
*<u>**Include all tools needed for each task with hazards and control measures**</u>*

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Identify the hazards associated with each step to complete the work

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Steps/Hazards/Control Measures

Steps to Complete Work	Hazards	Control Measures

Identify the various steps needed to complete each task from the tasks identified in section 2. ***Include all tools needed for each task with hazards and control measures***

Identify the hazards associated with each step to complete the work

Identify the Control Measures that are needed to mitigate the risks/hazards of each step

Adjacent Work/After Action Review

Work in Adjacent Areas:

ZERO INJURY OBLIGATION:
1. Conduct your work in a safe manner
2. Stop work immediately to correct any unsafe condition that is encountered
3. Take corrective action so that work may proceed in a safe manner

After Action Review	
What did not go according to plan today?	How can we prevent it in the future / fix it?

End of Shift Review
☐ Work area cleaned up (trash, tools, equipment, etc.)
☐ Permits closed out and turned in
☐ All injuries reported to supervisors

Emergency Contact:

Muster Area:

Work in Adjacent Areas:

Identify the other trades that are working next to / within your work area

Inform them of your work and hazards and engage in dialogue to what their work and hazards are

If necessary, inquire with Mortenson area Superintendent to determine who shall conduct their work first. Document their name and trade

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Emergency Contact:

Muster Area:

Make sure to input this information at the beginning of the shift in case of any emergency

Know your work area and the closest muster area and emergency route

At the end of the shift, reflect on the AAR questions and provide feedback if necessary

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Complete the End of Shift Review section when work is completed in the area or at the end of the shift. This is to ensure your crew has cleaned their work area so follow along work can start free from hazards

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Document their name and trade

What hazards do you see?

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Count the number of F’s

Count the number of F's

FINISHED FILES ARE THE RESULT OF YEARS OF SCIENTIFIC STUDY COMBINED WITH THE EXPERIENCE OF YEARS

Count the number of F’s

Count the number of F's

FINISHED FILES ARE THE RESULT OF YEARS OF SCIENTIFIC STUDY COMBINED WITH THE EXPERIENCE OF YEARS

(6)

Count the number of F’s

Count the number of F's

TWO OF THE MOST POWERFUL OF ALL HUMAN FEARS ARE THE FEAR OF FAILURE AND THE FEAR OF SUCCESS

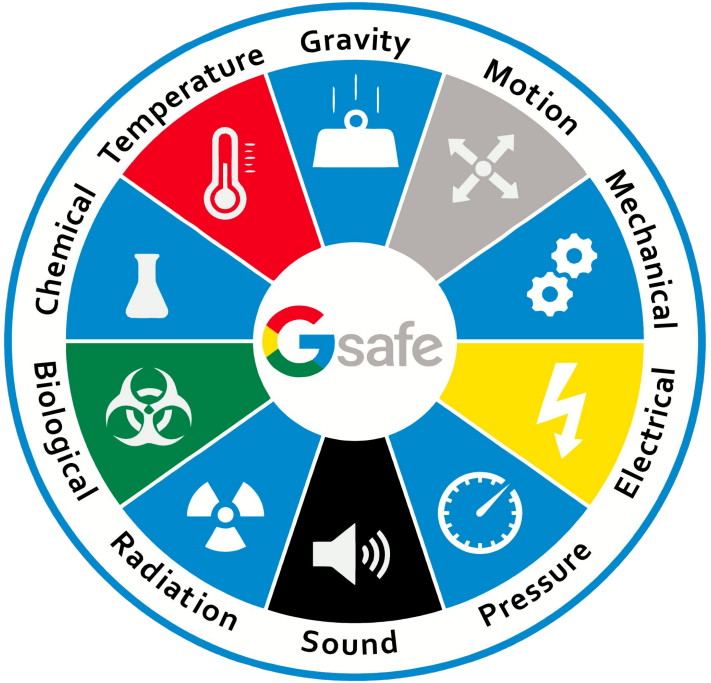
Count the number of F’s

Count the number of F's

TWO OF THE MOST POWERFUL OF ALL HUMAN FEARS ARE THE FEAR OF FAILURE AND THE FEAR OF SUCCESS

(9)

What hazards do you see ***NOW***?



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